Variation of the Geomagnetic Field Intensity Since the Late Miocene

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To estimate the variation of the geomagnetic field intensity since the late Miocene, the Thelliers stepwise heating method has been applied to a series of Neogene to Quaternary volcanic rocks taken from the Kuju volcanic region of Kyushu Island, west Japan. As a result we can conclude broadly that the geomagnetic field has generally increased in intensity from the late Miocene to the present, with fairly large and long-term fluctuations. It seems likely that the intensity of the geomagnetic field and its variation during the past 1 m.y. have not significantly differed from the intensity and variation for the past several thousand years.

INTRODUCTION

Until recently, most paleomagnetic studies have been focused on the directional changes of the ancient geomagnetic field. A few reports have been presented on the variation of the past geomagnetic field intensity [*Briden*, 1966; *Kono and Nagata*, 1966; *Coe*, 1967; *Smith*, 1967*a*, *b*; *Carmichael*, 1967]. *Coe* [1967], through his careful experiments, reported unusually weak intensity during a reversal in the Miocene; *Smith* [1967*a*] reported intensity estimates during the Tertiary period.

On the other hand, the suitability of igneous rocks for paleo-intensity determination was discussed by *Smith* [1967*b*] with particular reference to criteria for his modified method employing ac demagnetization. He also emphasized the difficulty of obtaining reliable specimens that give reproducibility of the ancient field intensity in heating experiments.

Thellier and Thellier's [1959] stepwise heating method is, however, considered to be one of the most rigorous methods for estimating the ancient field intensity. During heat treatments by their method, physicochemical changes sometimes occur in the ferromagnetic minerals contained in specimens; consequently, the experimental data must be discarded. For this reason, certain criteria for selecting proper specimens before intensity measurements are made are necessarily required, so that experimental work is not lost. The criteria described in a later section are applied in this study, although they are still insufficient to preclude every disadvantageous sample.

There is a series of Neogene to Quaternary lava flows in the Kuju volcanic region of Kyushu Island, west Japan. We intend to discuss primarily the relation between the remanent magnetic characteristics of those volcanic rocks and their suitability for the Thelliers intensity measurement and also discuss the approximate variation of the paleo-intensity ranging from upper Miocene to Recent.

MAGNETIC CHARACTERISTICS AND SAMPLE SELECTION

More than 200 oriented specimens were taken from twenty-one volcanic lava units of the Kuju volcanic region. The ages of these specimens estimated geologically range from upper Miocene to Recent [*Ono*, 1963].

The result of these paleomagnetic studies was published by *Nishida et al.* [1967]. The reliability tests of these NRMs were made by ac and thermal demagnetization in a nonmagnetic field. Thermomagnetic characteristics of the constituent ferromagnetic minerals of these rocks were measured in a strong ambient field of 3000 oe with a thermomagnetic quartz balance. The results are summarized in Table 1 and are also shown in Figure 1. Various magnetic properties, including site mean direction of NRM, Curie point of the major constituent, saturation moment at room temperature (J_s), intensity of NRM (J_n), and stability grade determined by the demagnetization tests, are given in Table 1. Additional minor components

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TABLE 1. Magnetic Properties and Rock Types of the Samples Used

Sample	Rock Type	Mean Direction		Curie Point, °C	J_s, emu/g	J_n, $\times 10^4$ emu/g	Stability
		D, deg	I, deg				
				Pleistocene			
A	Welded tuff	355.5	48.2	560	0.57	4.3	Unstable
Kp	Andesite	338.3	32.8	550	0.25	23.8	Stable
D4	Andesite	355.2	43.8	520	0.32	14.3	Stable
D2	Andesite	355.2	26.8	475	1.31	10.5	Unstable
K7	Andesite	323.7	34.4	545	0.13	53.8	Stable
K6	Andesite	350.4	51.6	450	1.14	9.2	Metastable
K4	Andesite	340.1	29.1	555	0.30	3.1	Stable
K2	Andesite	19.6	56.3	525	0.82	49.6	Unstable
Dh	Basalt	241.7	−51.2	525	1.20	16.4	Stable
				Pliocene			
H4	Andesite	358.6	65.5	540	1.38	2.8	Unstable
H2	Andesite	6.9	45.3	550	1.25	7.8	Unstable
H1	Andesite	348.7	32.6	490	0.89	11.8	Stable
Y	Rhyolite	357.6	46.0	500	0.13	3.9	Metastable
Rw	Rhyolite	177.5	−59.6	510	1.26	45.4	Stable
R1	Rhyolite	184.1	−41.0	530	0.06	12.6	Stable
P1(R)	Andesite	154.5	−40.3	535	0.70	40.3	Stable
P1(N)	Andesite	19.1	55.8	540	0.28	16.7	Unstable
I3	Welded tuff	345.9	37.0	530	0.13	0.6	Metastable
Tp	Andesite	5.4	55.7	525	0.70	6.9	Stable
				Upper Miocene			
03	Rhyolite	187.2	−72.3	530	0.65	51.4	Stable
01	Rhyolite	5.2	24.1	540	0.52	22.9	Unstable

with different Curie points were found in a few specimens.

The Thelliers stepwise heating method has been adopted in the present study in order to determine the ancient geomagnetic field intensity. The details of the experimental procedure were described in a previous paper [*Sasajima and Maenaka*, 1966], along with the variation of the ancient field intensity during historical and prehistoric ages in Japan. In that study, we used baked earths and potteries that are commonly considered to be the most appropriate samples for the Thelliers heating experiments. In general, volcanic rocks are likely to be sensitive to heat treatments, as compared with the baked earths. Therefore, special attention must be given to the selection of suitable samples for this method. Therefore, samples that seemed to be stable for heat treatments were selected according to the following criteria.

1. The thermomagnetic curve of a sample must be nearly reversible between the heating and the cooling processes. Furthermore, a single-phased sample is preferable to a poly-phased one. A change of 15% of initial magnetization after the cooling process is the maximum limit acceptable in this study.

2. The NRM of a sample must appear likely to be original TRM by ac or thermal demagnetization test. The maximum permissible limits of change in NRM are 20% in intensity and several degrees in direction under an alternating peak field of 150 oe or partial demagnetization to 150°C.

Result of Intensity Measurements

In the actual selection of samples, only the ones belonging to the 'stable' category in Table 1 were preferentially adopted. Five or six specimens from each sample were subjected to the intensity measurement. All but a few specimens of the preferred samples gave very good results. Samples Kp and K4 also gave fairly good results, although they had significantly different magnetizations before and after heating during the thermomagnetic analysis. Only a small num-

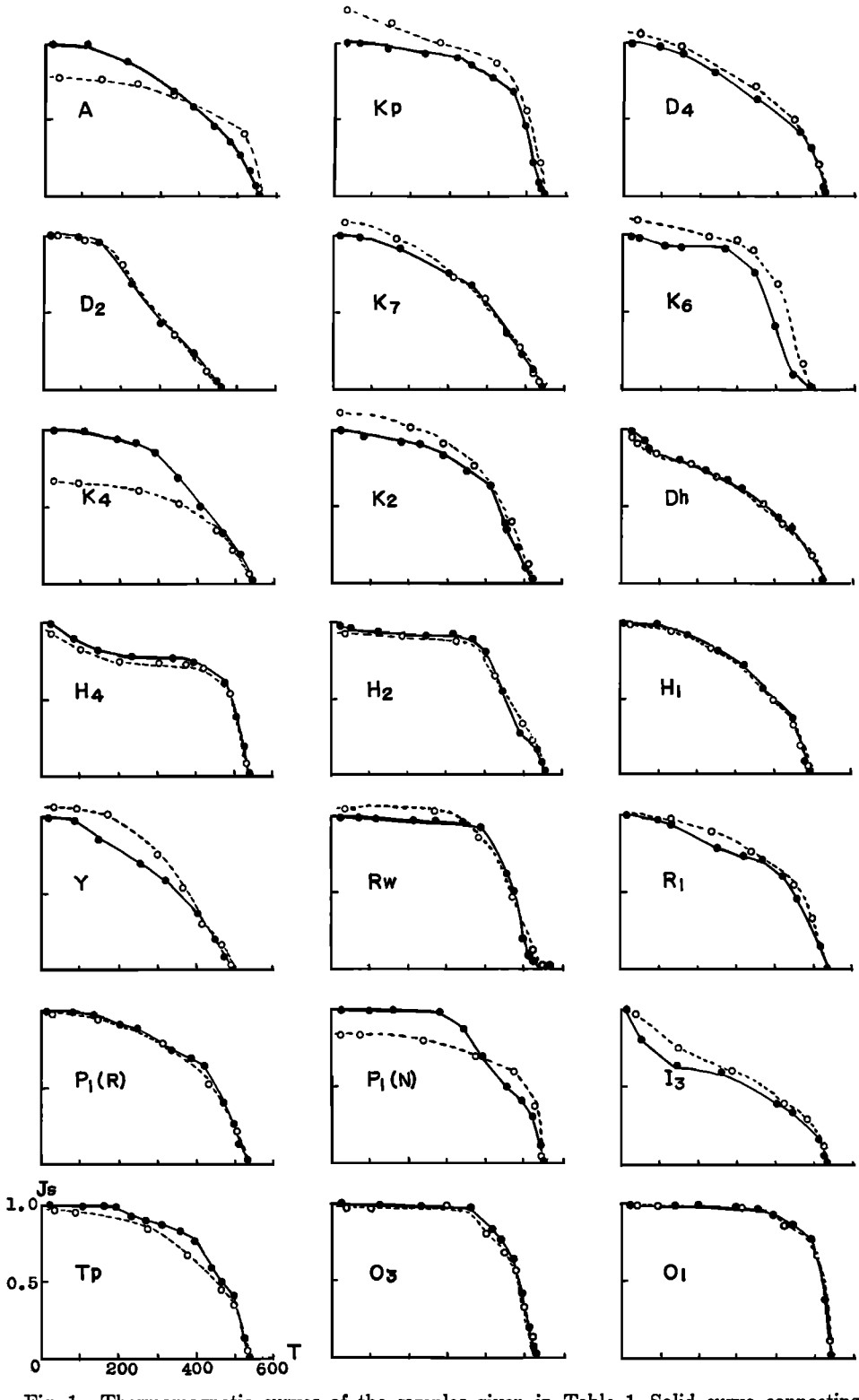
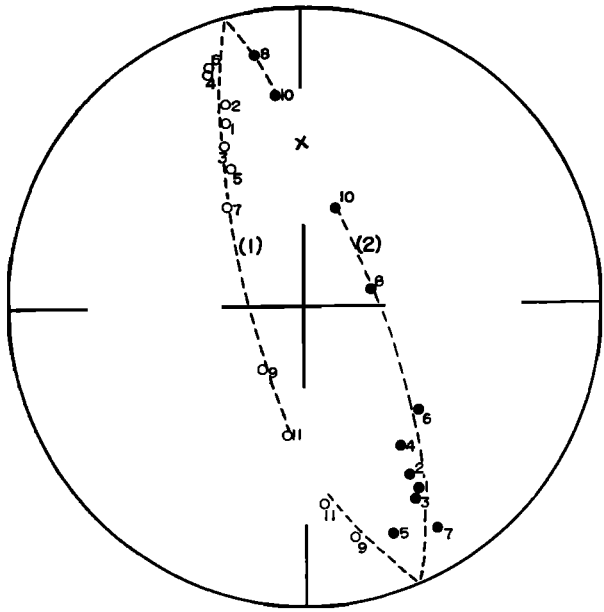
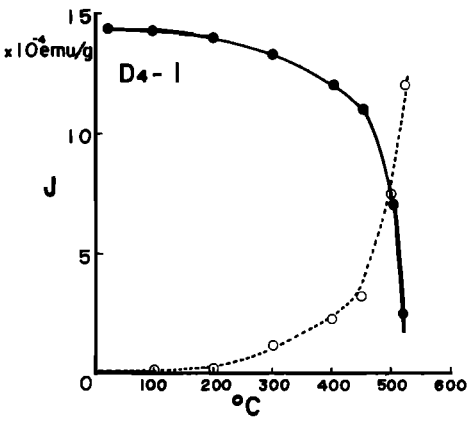
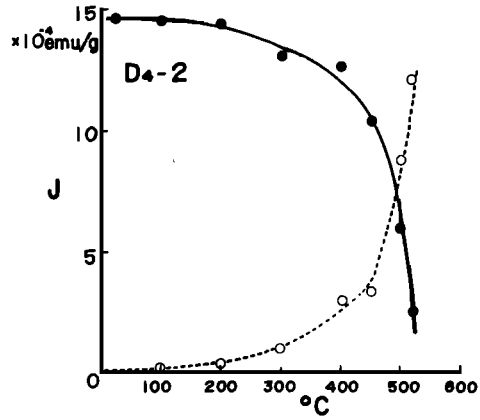
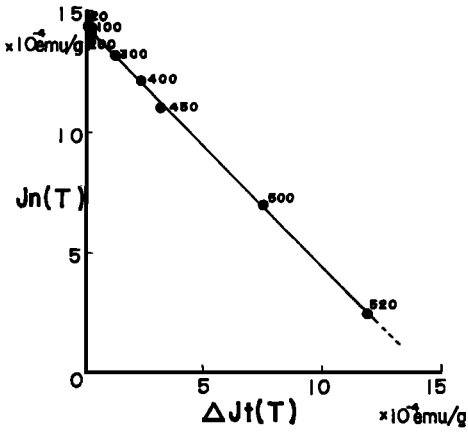


Fig. 1. Thermomagnetic curves of the samples given in Table 1. Solid curve connecting solid circles represents a heating process; dotted curve connecting open circles, a cooling process.


(1)
(2)


D4-1
J
°C


D4-2
J
°C


Jn(T)
ΔJt(T)

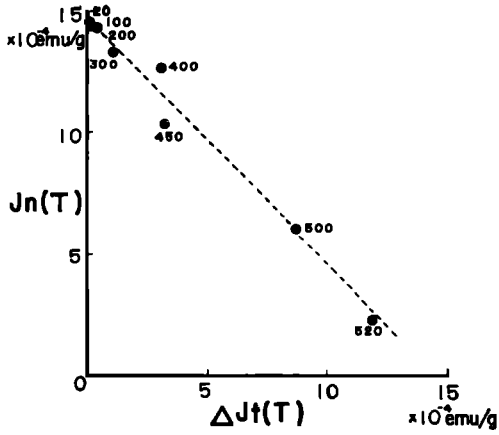

Jn(T)
ΔJt(T)

TABLE 2. Result of the Field Intensity Measurements and Corresponding Virtual Dipole Moment

Sample	Polarity	No.	Measured F/F_0	Mean F/F_0	F, oe	VDM, $\times 10^{-25}$ oe cm^3
			Pleistocene			
Kp	N	2	0.45, 0.47	0.46 ± 0.01*	0.215	4.57
D4	N	4	0.87, 0.95, 0.99, 1.40	1.05 ± 0.23	0.491	10.44
K7	N	5	0.82, 0.83, 0.92, 1.03, 1.11	0.94 ± 0.13	0.439	9.33
K4	N	3	1.25, 1.31, 1.50	1.35 ± 0.13	0.631	13.42
Dh	R	4	0.94, 1.03, 1.03, 1.26	1.06 ± 0.14	0.496	10.55
			Average (18)	1.01 ± 0.28	0.472	9.66
			Pliocene			
H1	N	1	0.59	0.59	0.276	5.42
Rw	R	3	0.68, 0.72, 0.92	0.77 ± 0.13	0.360	7.07
R1	R	1	0.88	0.88	0.411	8.08
P1	R	6	0.73, 0.77, 0.84, 0.87, 0.98, 1.02	0.87 ± 0.11	0.406	7.98
Tp	N	3	0.89, 1.00, 1.13	1.00 ± 0.12	0.467	9.18
			Average (14)	0.86 ± 0.15	0.401	7.55
			Upper Miocene			
03	R	3	0.53, 0.56, 0.72	0.60 ± 0.09	0.280	5.52
			Average (3)	0.60 ± 0.09	0.280	5.52

* Standard deviation of mean F/F_0.

ber of these specimens failed to meet the selection criteria. On the other hand, only one specimen was acceptable from each of samples H1 and R1, for which thermomagnetic curves were assumed doubtful whether they were single-phased or not.

Judging from the J_s-T curves of Figure 1 and the magnetic characteristics given in Table 1, sample D4 was assumed to be suitable for the paleo-intensity measurement; the result of intensity measurement on this sample is illustrated in Figure 2. In the upper diagram of the figure, the vector changes of remanence under a sequence of experiments are drawn on a Schmidt's net. Whether or not an experiment is working successfully can also be checked by a directional change on the diagram. If the original NRM could be exclusively attributable to TRM, all the directions of Schmidt's net, which are given from both normal and reverse settings, should evidence a regular movement along the great circle including the ambient field direction in the laboratory. The middle diagram in Figure 2 illustrates a relation between the progressive demagnetization of NRM (solid line) and the acquisition of TRM (broken line) of the specimen in the present earth's field. In the lower diagram, $\Delta J_n(T)$ versus $J_t(T)$ is shown, where $\Delta J_n(T)$ denotes a residual NRM intensity under progressive demagnetization at temperature T°C and $J_t(T)$ is the partial TRM of the same specimen acquired at T°C in the present geomagnetic field. These points must fall on a straight line, according to the Thelliers principle, provided that the NRM of a specimen is completely attributable to its TRM acquired at the

Fig. 2. Result of an experimental run of the Thelliers method that was applied in the present study. In the upper diagram solid circles are in lower hemisphere and open circles are in the upper hemisphere. The numerals near the circles represent the order of successive experiments. The temperatures are as follows: 1, 20°–200°C; 2 and 3, 300°C; 4 and 5, 400°C; 6 and 7, 450°C; 8 and 9, 500°C; 10 and 11, 520°C. Cross denotes the direction of ambient field in the laboratory. The numerals by circles of the lowest diagram denote the temperature in degrees Celsius. See text for further explanation.

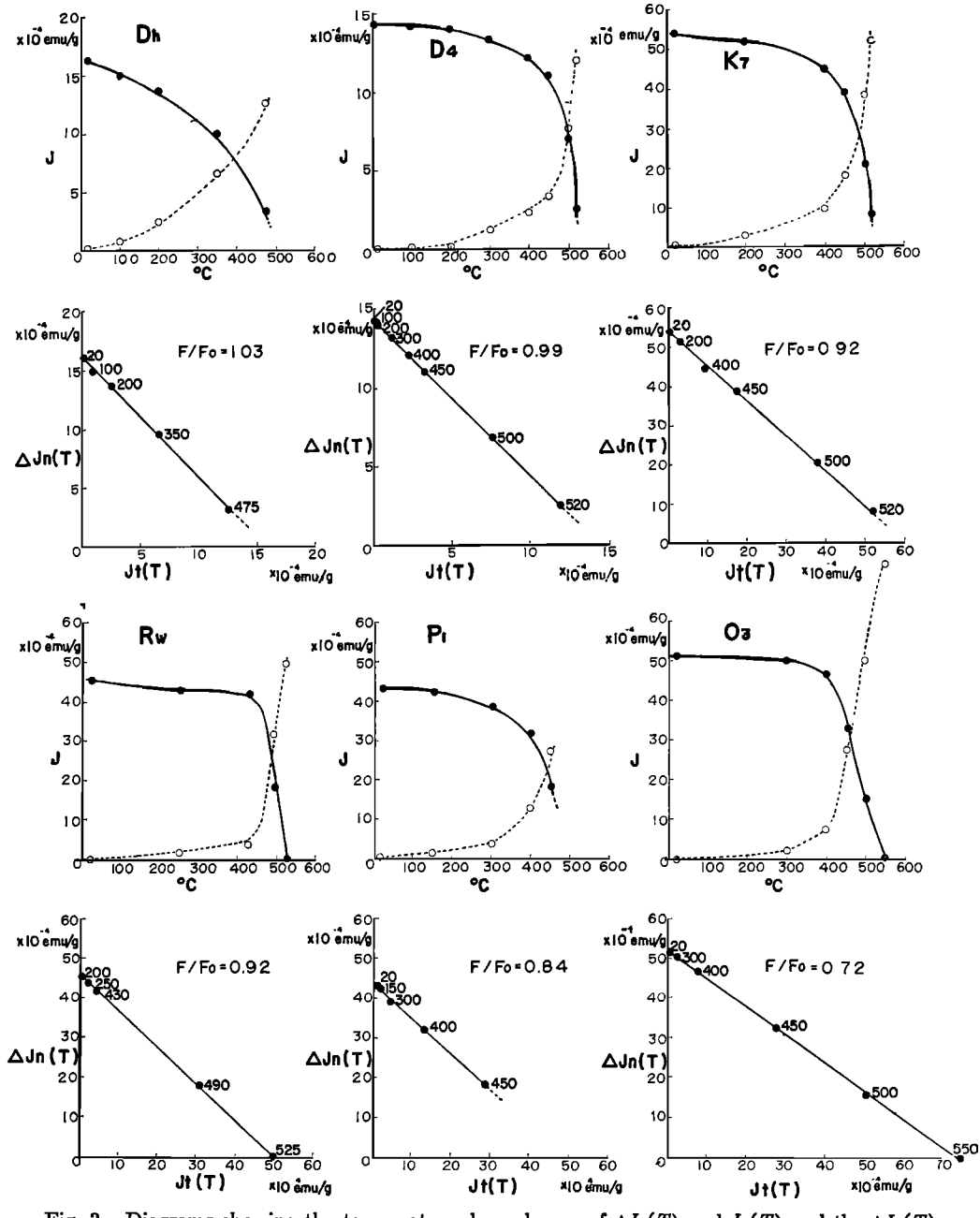


Fig. 3. Diagrams showing the temperature dependences of $\Delta J_n(T)$ and $J_t(T)$ and the $\Delta J_n(T)$ versus $J_t(T)$ relation of suitable samples.

time of initial cooling and provided that any appreciable change in ferromagnetic minerals did not occur during experimental heating. The gradient of the straight line gives an intensity ratio F/F_0, where F is the ancient geomagnetic field intensity at the sampling site and F_0 is the present intensity in the laboratory (0.467 oe in Kyoto). In this diagram, the fairly well defined linear relation of specimen D4-2 is in reasonable conformity with the directional result shown

in the upper diagram. By comparing the directional change in the upper diagram with its corresponding intensity change shown in the lower one, it is apparent that the accuracy of specimen D4-2 is worse than the accuracy of D4-1. A result such as that for D4-2 is excluded from the intensity data given in Table 2. Other examples of successful results are given in Figure 3.

The result of the intensity measurements is expressed in terms of F/F_0 in Table 2 and is tentatively drawn in Figure 4. The abscissa of Figure 4 defines the age of the sample, which has been estimated from geological evidence and the paleomagnetic correlation based on the geomagnetic polarity time scale so far available [*Dalrymple et al.*, 1967].

Discussion and Conclusions

As already mentioned, the Thelliers heating method has been shown, with reasonable certainty, to be applicable to various kinds of volcanic rocks. This conclusion should, however, be taken with the reservation that about one-fifth of the specimens measured in the present study have been discarded from the final result because of the physicochemical change of ferromagnetic minerals during heat treatment. In this regard, *Coe* [1967] and *Kono and Nagata* [1966] were independently obliged to reject about one-third of the data obtained from a variety of volcanic rocks taken from the western United States. Therefore, it is likely that the criteria for pre-selection of samples must have played an effective role, as judged by the improved rate of success in the present study.

From an examination of Table 2 and Figure 4, it should be emphasized that comparatively large spreads in intensity values, namely, a variation by a factor of at least 3, might have occurred within less than the past 1 m.y., if estimated ages of samples have been correct. This variation by a factor of 3 is nearly consistent with the variation of the past several thousand years found in the study of archaeomagnetism.

Another important fact to be mentioned is that the intensity ratios obtained from several specimens cut from a single fist-sized sample give considerably different values. From this fact it may be suggested that several specimens at least are needed to determine a reliable field intensity value from even a single lava flow.

Recently *Smith* [1967*a*] proposed a virtual dipole moment (VDM), which is analogous to the virtual geomagnetic pole position (VGP) calculated from mean direction of NRMs at any site. By making use of VDM, field intensity data given at different sampling places in the world can be compared directly with one another. In this study, calculation of VDM is performed by connecting the intensity ratio

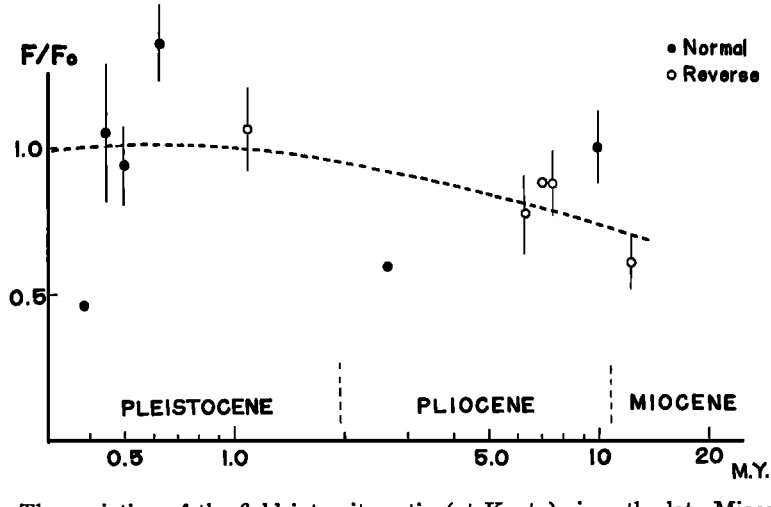


Fig. 4. The variation of the field intensity ratio (at Kyoto) since the late Miocene. Short line on a plotted circle represents standard deviation of the mean value. The general trend of variation is drawn by a thick dotted line.

TABLE 3. Summary of Mean Virtual Dipole Moments

	VDM, $\times 10^{-25}$ oe cm^3	Standard Deviation, $\times 10^{-25}$ oe cm^3	Standard Error, $\times 10^{-25}$ oe cm^3
Pleistocene	9.66	2.87 (29.7%)	1.28 (13.3%)
Pliocene	7.55	1.25 (16.5%)	0.56 (7.5%)
Upper Miocene	5.52	...	...
Upper Neogene	7.21	1.37 (19.0%)	0.56 (7.8%)
Average of 6 results with normal polarity	8.73	2.99 (34.2%)	1.22 (14.0%)
Average of 5 results with reversed polarity	7.84	1.64 (21.0%)	0.73 (9.3%)
Average of all 11 data	8.32	2.51 (30.1%)	0.76 (9.1%)

given in Table 2 with the mean direction of NRM given in Table 1. The computed values of the VDM are listed in the last column of Table 2. Mean VDM for Pleistocene, Pliocene, and upper Neogene (including upper Miocene) are given in Table 3, together with various kinds of mean VDM.

From the result in the table, it may be noted that (1) on the average, geomagnetic field intensity did not significantly change during different polarity epochs, although relatively larger fluctuations were included in normal data compared with the fluctuations in reversed data, and (2) the VDM has increased gradually, as a whole, from upper Miocene to the present, accompanying relatively long-term fluctuations superimposed on the general trend. The second result may roughly conform with the result obtained by *Coe* [1967] from rocks of the western United States and also by *Smith* [1967*a*] from Icelandic baked laterites.

The averaged VDM of the upper Neogene found by us, 7.21×10^{25} oe cm^3 is comparatively larger than the VDM reported by *Coe* [1967] and *Smith* [1967*a*]. Coe's value is about 0.6 times the present value, and Smith's value is 5.34×10^{25} oe cm^3 for almost the same age.

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